

LAB # 13

Sorting Elements in an Array using MIPS Assembly Language

Objective:

To implement a sorting operation in MIPS assembly language that arranges elements of an array in a specified order using loops and comparison instructions.

Theory:

Sorting is a basic operation performed on arrays. In MIPS, arrays are handled using memory addresses, loop counters, and comparison instructions.

Sorting:

Sorting is the process of arranging elements of an array in a particular order, usually ascending (smallest to largest) or descending (largest to smallest). In MIPS assembly language, sorting can be implemented using loops, comparison instructions, and conditional branching.

For Sorting We Implement in MIPS:

- Loading the base address of the array
- Reading elements using lw
- Using loops to traverse the array
- Comparing values using branch instructions
- Swapping elements when necessary using temporary registers
- Printing the sorted array using system calls

Sample Program:

```
.data
arr: .word 1,6,2
n:   .word 3
msg: .asciiz "Sorted Array"
space: .asciiz " "
```

```

.text
.globl main
main:

    la $t0, arr    # array
start
    lw $t1, n      # size

outer:
    li $t2, 0      # index j
start
    la $t3, arr    # pointer
reset

inner:
    addi $t4, $t2, 1
    bge $t4, $t1, next_outer

    lw $s0, 0($t3) # current
    lw $s1, 4($t3) # next

    ble $s0, $s1, skip
    sw $s1, 0($t3) # swap
    sw $s0, 4($t3)

skip:
    addi $t3, $t3, 4
    addi $t2, $t2, 1
    j inner

next_outer:
    li $t2, 0      # reset inner index
    addi $t1, $t1, -1 # reduce loop end for optimization
    bgt $t1, 1, outer # repeat until only 1 element left

exit:
    li $v0, 10
    syscall

```

Output:

```
Sorted Array: 1 2 6  
-- program is finished running --
```

LAB TASKS:

1. Write a MIPS program that takes an array of 12 integers as input from the user, sorts it in ascending order, and displays the sorted array.
2. Write a MIPS program that sorts a given array of integers in descending order and displays the result.

Arr: 3,8,9,2,7,6,4,1,5,10